

SOFTWARE ENGINEERING · REQUIREMENTS AND PROTOTYPING

CUSports Sprint 0 Worksheets

Worksheet Order – Each Worksheet maps to a Step in the Story 0 Prompt

0. Team information and AI-evidence index
1. Initial AI-produced stories and acceptance tests (unedited)
2. Revision log
3. Prioritized backlog
4. Dependency identification
5. Prototype plan
6. Prototype walkthrough results
7. Final stories and acceptance tests + Change analysis

WORKSHEET 0**Team record, prompt plan, and AI evidence index****Team Name and Number: Team 4 - Fantastic Four****Product Context and Roles**

CUSports is an online Clemson sporting-goods retailer. Its product experience must support accurate product availability, ordering, warehouse work, shipping, customer support, system operation, and business reporting. Your team will turn this broad context into a coherent, testable product backlog and a supporting prototype.

CODE	ROLE	CODE	ROLE
CUS	Customer	ADM	System administrator
WHS	Warehouse employee	CSR	Customer service representative
SHP	Shipper	BIZ	Business user

Initial-generation prompt plan

PROMPT ELEMENT	TEAM PLAN
Product context to give AI	CUSports is an online Clemson sporting goods retailer. It supports accurate product availability, ordering, warehouse work, shipping, customer support, system operation, and business reporting.
Roles and coverage	The six roles include customers, system admins, warehouse employees, customer service reps, shippers, and business users
Requested output	There should be 5 user stories per role, in user story form (Who, What, Why), each story should have 4 acceptance tests. User stories should follow the id system of US-01, US-02, ... and acceptance tests should follow AT-01a, AT-01B, AT-01C, AT-01D, AT-02a, AT-02B, AT-02C, AT-02D,
Quality constraints	Each user story should include easily observable tests, clearly state any assumptions about the users and should not invent requirements, include edge cases that are relevant to the specific story, keep each story small enough to be implemented and tested on its own and do not expand scope based on assumptions.
Information intentionally withheld or unknown	When an unknown detail could change implementation or acceptance criteria, flag it as a question instead of making an assumption.

AI evidence index

ID	STEP	TOOL / MODEL	PURPOSE	LOCATION
A1	Step 1	ChatGPT	Generate User Stories	Pg 29 - Pg 37, pg 40 - pg 47
A2	Step 1	Claude	Generate User Stories	Pg 37- Pg40
A3	Step 5	Claude	Generate the prototype	FantasticFour_Sprint0.zip
A4	[Step 1 or 5]	[Tool/model]	[Purpose]	[Evidence page/file]
A5	[Step 1 or 5]	[Tool/model]	[Purpose]	[Evidence page/file]
A6	[Step 1 or 5]	[Tool/model]	[Purpose]	[Evidence page/file]

AI-use attestation

Your submission of this worksheet affirms that AI was used only for initial user-story/acceptance-test generation and prototype creation; all prompts and results are included; and prioritization, dependency analysis, story/test refinement, rationale, and reflection are our own work.

WORKSHEET 1

Initial AI-produced backlog

ID / SOURCE	ROLE	INITIAL STORY — VERBATIM	INITIAL ACCEPTANCE TESTS — VERBATIM
US-01 AI ID: A1	Customer	Who: Customer What: I want to view whether a sporting goods product is available so that I can determine whether it can be ordered.	AT-01a: Given a product with availability information, when the customer views the product, then the system displays the product's current availability status. AT-01B: Given a product that is not available, when the customer views the product, then the system does not indicate that the product is available. AT-01C: Given a product whose availability information cannot be determined, when the customer views the product, then the system does not incorrectly indicate that the product is available. AT-01D: Given a customer views a product, when the availability status is displayed, then the status corresponds to the availability information maintained by the system.
US-02 AI ID: A1	Customer	Who: Customer What: I want to order available sporting goods so that I can purchase products from CUSports.	AT-02a: Given an order containing available product(s), when the customer submits the order, then the system creates the order. AT-02B: Given a product that is unavailable, when the customer adds the product to the order, then the system does not add the unavailable product to the order. AT-02C: Given an order containing multiple products, when one required product cannot be ordered, then the system does not silently create an order containing a different set of products. AT-02D: Given a successfully submitted order, when the order is created, then the system associates the ordered product(s) with that order.
US-03 AI ID: A1	Customer	Who: Customer What: I want to view information about my order so that I can understand what I ordered.	AT-03a: Given a customer has an order, when the customer accesses that order, then the system displays information identifying the order. AT-03B: Given an order contains one or more products, when the customer views the order, then the system displays the products associated with that order. AT-03C: Given a customer has no orders, when the customer accesses their orders, then the system does not display another customer's order as their own. AT-03D: Given an order exists, when the customer views it, then the displayed order information reflects the information stored for that order.
US-04 AI ID: A1	Customer	Who: Customer What: I want to access shipping information for my order so that I can understand the status of its shipment.	AT-04a: Given an order has shipping information, when the customer views the order, then the system displays the available shipping information. AT-04B: Given an order does not yet have shipping information, when the customer views the order, then the system does not display fabricated shipping information. AT-04C: Given shipping information changes, when the customer views the order after the change has been recorded, then the displayed information reflects the recorded information. AT-04D: Given a customer has multiple orders, when the customer views shipping information for one order, then the system displays shipping information associated with that order rather than another order.

ID / SOURCE	ROLE	INITIAL STORY — VERBATIM	INITIAL ACCEPTANCE TESTS — VERBATIM
US-05 AI ID: A1	Customer	Who: Customer What: I want to contact CUSports customer support so that I can request assistance with my order or account.	AT-05a: Given a customer has access to the customer-support mechanism, when the customer submits a support request, then the system records the submitted request. AT-05B: Given a customer submits a support request, when required information is missing, then the system does not record the request as complete. AT-05C: Given a customer submits a support request, when the submission succeeds, then the system provides an observable indication that the request was submitted. AT-05D: Given a customer submits a support request concerning an order, when the request is recorded, then the order information supplied by the customer remains associated with the request.
US-06 AI ID: A1	System Admin	Who: System Administrator What: I want to manage system-user access so that authorized users can operate the CUSports system.	AT-06a: Given the administrator has access to user-management functionality, when the administrator views system users, then the system displays the users available for management. AT-06B: Given the administrator performs a supported user-management action, when the action succeeds, then the resulting user information reflects that action. AT-06C: Given the administrator attempts to perform an unsupported or invalid user-management action, then the system does not apply the invalid change. AT-06D: Given a user-management change is successfully made, when the administrator views the affected user afterward, then the displayed information reflects the saved change.
US-07 AI ID: A1	System Admin	Who: System Administrator What: I want to maintain product information so that CUSports can provide accurate product data.	AT-07a: Given an administrator has access to product management, when the administrator views a product, then the system displays the product information stored for that product. AT-07B: Given the administrator makes a supported change to product information, when the change is saved successfully, then the system displays the updated information. AT-07C: Given the administrator attempts to save invalid or incomplete product information where validation is required, then the system does not save the invalid information. AT-07D: Given an administrator changes product information, when a customer subsequently views that product, then the customer sees the product information currently maintained by the system.
US-08 AI ID: A1	System Admin	Who: System Administrator What: I want to maintain product availability so that customers receive accurate availability information and can order products that are available.	AT-08a: Given an administrator has access to availability management, when the administrator records a supported availability change, then the system saves the new availability information. AT-08B: Given a product is marked unavailable, when a customer attempts to order that product, then the system does not treat the product as available. AT-08C: Given a product's availability is changed from unavailable to available, when the change is successfully saved, then the system uses the updated availability information. AT-08D: Given an administrator attempts to save an invalid availability value, then the system does not save the invalid value.

ID / SOURCE	ROLE	INITIAL STORY — VERBATIM	INITIAL ACCEPTANCE TESTS — VERBATIM
US-09 AI ID: A1	System Admin	Who: System Administrator What: I want to maintain the system information needed for warehouse and shipping operations so that orders can be processed and shipped accurately.	AT-09a: Given an order exists in the system, when an authorized warehouse or shipping operation updates supported order information, then the system records the update. AT-09B: Given an order does not exist, when an operation attempts to update that order, then the system does not create or update an unrelated order. AT-09C: Given required information for a supported warehouse or shipping update is missing, when the update is submitted, then the system does not save the incomplete update. AT-09D: Given a warehouse or shipping update is successfully recorded, when the order is subsequently viewed, then the system displays the recorded information.
US-10 AI ID: A1	System Admin	Who: System Administrator What: I want to generate business reports so that I can review information about CUSports operations.	AT-10a: Given the administrator has access to a supported report, when the administrator requests the report, then the system generates the report using the available system data. AT-10B: Given the requested report has no matching data, when the administrator generates the report, then the system indicates that no matching data is available rather than displaying unrelated data. AT-10C: Given a report-generation request contains an invalid or unsupported parameter, then the system does not generate a report using the invalid parameter as though it were valid. AT-10D: Given a report is successfully generated, when the administrator views it, then the report reflects the underlying system data used to generate it.
US-11 AI ID:A2	Business User	Who: Business User What: I want to view a sales report for a date range I select, so that I can understand revenue and order volume performance over a specific period.	AT-11a: Given a business user is logged in and on the reporting section, when they select a valid start date and end date and submit, then a report displays total number of orders and total sales revenue for that exact range. AT-11B: Given a business user enters a start date that is after the end date, when they submit, then the system displays a validation error and no report is generated. AT-11C: Given a business user selects a date range with zero completed orders, when they submit, then the report displays zero orders and \$0.00 revenue rather than an error or blank screen. AT-11D: Given a business user selects a date range spanning more than one calendar year, when they submit, then the report still returns accurate totals for the full selected range (no silent truncation).
US-12 AI ID:A2	Business User	Who: Business User What: I want to view a report of products at or below a defined stock threshold, so that I can identify items that may need reordering.	AT-12a: Given products exist with stock quantities below the defined threshold, when the business user opens the low-stock report, then all such products are listed with their product name and current quantity. AT-12B: Given no products are currently below the threshold, when the business user opens the report, then the system displays a clear "no low-stock items" message rather than an empty/blank table. AT-12C: Given a product's stock quantity changes after the report is generated, when the business user refreshes/regenerates the report, then the updated quantity is reflected. AT-12D: Given a product has a stock quantity of exactly zero, when the business user views the report, then that product appears in the report (zero counts as below/at threshold).

ID / SOURCE	ROLE	INITIAL STORY — VERBATIM	INITIAL ACCEPTANCE TESTS — VERBATIM
US-13 AI ID:A2	Business User	Who: Business User What: I want to export a generated report to a downloadable file, so that I can share or analyze the data outside the system.	AT-13a: Given a report is displayed on screen, when the business user selects "Export," then a file downloads containing the same data shown on screen. AT-13B: Given a report contains zero rows of data, when the business user exports it, then the downloaded file is still generated successfully with headers only (no error/crash). AT-13C: Given an export is in progress, when the business user attempts to trigger a second export before the first completes, then the system either queues, blocks, or clearly indicates the export is already running (no duplicate/corrupted files). AT-13D: Given the exported file is opened, when the business user inspects it, then column headers match the on-screen report labels exactly.
US-14 AI ID:A2	Business User	Who: Business User What: I want to view a ranked list of top-selling products over a date range, so that I can identify which products are driving the most business.	AT-14a: Given completed orders exist within a selected date range, when the business user requests the top-selling products report, then products are listed in descending order by units sold. AT-14B: Given two or more products have the exact same units sold, when the report is generated, then both products appear in the report without one being dropped or overwritten. AT-14C: Given a selected date range has no completed orders, when the report is generated, then the system displays a clear empty-state message instead of an error. AT-14D: Given a product was sold during the range but has since been removed/discontinued from the catalog, when the report is generated, then it still appears in historical results (report reflects sales history, not current catalog status).
US-15 AI ID:A2	Business User	Who: Business User What: I want to view a report showing the number of customer support tickets opened and resolved over a date range, so that I can monitor customer support workload and performance trends.	AT-15a: Given tickets were opened and/or resolved within a selected date range, when the business user generates the report, then it shows separate counts for tickets opened and tickets resolved in that range. AT-15B: Given a ticket was opened before the selected range but resolved within it, when the report is generated, then it is correctly counted only in the "resolved" total for that range, not "opened." AT-15C: Given no tickets were opened or resolved in the selected range, when the report is generated, then it displays zero counts rather than an error or blank output. AT-15D: Given a ticket is still open/unresolved as of the report date, when the report is generated, then it is excluded from the "resolved" count but included in the "opened" count if applicable to the range.
US-16 AI ID:A1	Shipper	User story: As a shipper , I want to view orders that are ready for shipment , so that I can identify which orders need to be processed .	AT-16a: Given an order meets the system's existing criteria for being ready to ship, when the shipper views the ready-to-ship orders, then that order is displayed. AT-16b: Given an order does not meet the criteria for being ready to ship, when the shipper views the ready-to-ship orders, then that order is not displayed. AT-16c: Given there are no orders currently ready to ship, when the shipper views the list, then the system displays an observable indication that there are no ready-to-ship orders. AT-16d: Given multiple orders are ready to ship, when the shipper views the list, then each qualifying order is displayed as a distinct order that the shipper can identify.

ID / SOURCE	ROLE	INITIAL STORY — VERBATIM	INITIAL ACCEPTANCE TESTS — VERBATIM
US-17 AI ID:A1	Shipper	As a shipper , I want to view the information needed to prepare an order for shipment , so that I can process the correct order and send it to the correct destination .	AT-17a: Given the shipper selects an order available for shipment, when the order details are opened, then the system displays the shipment information defined as required for that order. AT-17b: Given the selected order contains multiple products, when the shipper opens its details, then each product included in the order is displayed. AT-17c: Given an order has incomplete required shipment information, when the shipper opens the order, then the system identifies the missing information rather than presenting the order as having complete shipment information. AT-17d: Given the shipper is viewing one order, when the shipper opens a different order, then the displayed shipment information corresponds to the newly selected order.
US-18 AI ID:A1	Shipper	As a shipper , I want to record that an order has been shipped , so that CUSports has an up-to-date record of the order's shipping status .	AT-18a: Given an order is eligible to be marked as shipped, when the shipper records the shipment as complete, then the order's status changes to the system's defined shipped status. AT-18b: Given an order has already been marked as shipped, when the shipper attempts to mark it as shipped again, then the system does not create a second shipment completion for the same order. AT-18c: Given an order is not eligible to be marked as shipped, when the shipper attempts to record shipment completion, then the system does not mark the order as shipped. AT-18d: Given the shipment is successfully recorded, when the shipper returns to the order, then the order displays its updated shipped status.
US-19 AI ID:A1	Shipper	As a shipper , I want to identify orders that cannot currently be shipped , so that I can avoid incorrectly completing their shipment .	AT-19a: Given an order has a system-recognized shipment-blocking condition, when the shipper views the order, then the system indicates that the order cannot currently be shipped. AT-19b: Given an order has no system-recognized shipment-blocking condition, when the shipper views the order, then the system does not identify it as having a shipment exception. AT-19c: Given an order has a shipment exception, when the shipper attempts to complete shipment, then the system prevents the shipment from being recorded as completed. AT-19d: Given the shipment-blocking condition is removed, when the shipper subsequently views the order, then the order no longer shows that exception if the system's rules determine it is now eligible for shipment.
US-20 AI ID:A1	Shipper	As a shipper , I want to view the current shipping status of an order , so that I can determine whether the order still requires shipping work .	AT-20a: Given an order has a recorded shipping status, when the shipper views the order, then the current status is displayed. AT-20b: Given an order's shipping status has been updated, when the shipper views the order after the update, then the newly recorded status is displayed. AT-20c: Given an order has no valid shipping status, when the shipper views the order, then the system does not display an incorrect status and instead indicates that the status is unavailable or invalid. AT-20d: Given two orders have different shipping statuses, when the shipper views each order, then each order displays its own corresponding status.

ID / SOURCE	ROLE	INITIAL STORY — VERBATIM	INITIAL ACCEPTANCE TESTS — VERBATIM
US-21 AI ID: A1	Customer Service Rep	User Story: As a customer service representative, I want to view the current status of a customer's order so that I can accurately answer customer questions about their order.	AT-21A: GIVEN a valid order exists, WHEN the CSR opens the order, THEN the system displays the order's current status. AT-21B: GIVEN an order cannot be found, WHEN the CSR searches for it, THEN the system indicates that no matching order was found. AT-21C: GIVEN an order's status changes, WHEN the CSR views the order afterward, THEN the displayed status reflects the updated status.
US-22 AI ID: A1	Customer Service Rep	User Story: As a customer service representative, I want to view the products included in a customer's order so that I can answer questions about what the customer purchased.	AT-22A: GIVEN a valid order exists, WHEN the CSR views its details, THEN the system displays the products associated with the order. AT-22B: GIVEN an order contains multiple products, WHEN the CSR views the order, THEN each ordered product is shown separately. AT-22C: GIVEN an order has a recorded quantity for a product, WHEN the CSR views that product in the order, THEN the correct ordered quantity is displayed.
US-23 AI ID: A1	Customer Service Rep	User Story: As a customer service representative, I want to check whether a product is currently available so that I can provide customers with accurate availability information.	AT-23A: GIVEN an available product, WHEN the CSR checks its availability, THEN the system reports that the product is available. AT-23B: GIVEN an unavailable product, WHEN the CSR checks its availability, THEN the system reports that the product is unavailable. AT-23C: GIVEN product availability changes, WHEN the CSR checks the product afterward, THEN the current availability is displayed.
US-24 AI ID: A1	Customer Service Rep	User Story: As a customer service representative, I want to view the shipping status of an order so that I can answer customer questions about its shipment.	AT-24A: GIVEN an order has entered the shipping process, WHEN the CSR views the order, THEN the system displays its current shipping status. AT-24B: GIVEN an order has not entered the shipping process, WHEN the CSR views shipping information, THEN the system indicates that shipping information is not yet available. AT-24C: GIVEN the shipping status has changed, WHEN the CSR views the order afterward, THEN the updated shipping status is displayed.
US-25 AI ID: A1	Customer Service Rep	User Story: As a customer service representative, I want to locate a customer order using its order identifier so that I can access the correct order while assisting the customer.	AT-25A: GIVEN a valid order identifier, WHEN the CSR searches for it, THEN the matching order is displayed. AT-25B: GIVEN an identifier that does not match an order, WHEN the CSR searches for it, THEN the system reports that no matching order was found. AT-25C: GIVEN multiple orders exist in the system, WHEN the CSR searches using a specific valid order identifier, THEN only the matching order is returned.

ID / SOURCE	ROLE	INITIAL STORY — VERBATIM	INITIAL ACCEPTANCE TESTS — VERBATIM
US-26 AI ID: A1	Warehouse Employee	User Story: As a warehouse employee, I want to view orders that require warehouse work so that I know which orders need to be prepared.	AT-26A: GIVEN orders are awaiting warehouse work, WHEN the warehouse employee views the work list, THEN those orders are displayed. AT-26B: GIVEN an order no longer requires warehouse work, WHEN the work list is refreshed, THEN that order is no longer shown as awaiting work. AT-26C: GIVEN no orders currently require warehouse work, WHEN the employee views the work list, THEN the system indicates that no orders are awaiting warehouse work.
US-27 AI ID: A1	Warehouse Employee	User Story: As a warehouse employee, I want to view the products and quantities required for an order so that I can prepare the correct items.	AT-27A: GIVEN a valid warehouse order, WHEN the employee opens it, THEN the required products are displayed. AT-27B: GIVEN a product has a required quantity greater than one, WHEN the employee views the order, THEN the required quantity is displayed. AT-27C: GIVEN an order contains multiple products, WHEN the employee opens it, THEN all products associated with that order are shown.
US-28 AI ID: A1	Warehouse Employee	User Story: As a warehouse employee, I want to record when required products have been picked so that the system reflects the progress of warehouse work.	AT-28A: GIVEN a required product has not been picked, WHEN the employee confirms the product was picked, THEN the system records it as picked. AT-28B: GIVEN multiple products are required for an order, WHEN one product is confirmed as picked, THEN the remaining products remain unconfirmed. AT-28C: GIVEN a product has already been confirmed as picked, WHEN the order is viewed again, THEN the product remains recorded as picked.
US-29 AI ID: A1	Warehouse Employee	User Story: As a warehouse employee, I want to report when a required product is unavailable so that the order is not incorrectly treated as ready for further processing.	AT-29A: GIVEN a required product cannot be located as available, WHEN the employee reports it as unavailable, THEN the system records the product as unavailable for that order. AT-29B: GIVEN a product has been reported unavailable, WHEN the order is viewed again, THEN its unavailable status is displayed. AT-29C: GIVEN an order contains an unavailable required product, WHEN warehouse progress is viewed, THEN the order is not shown as having all required products successfully prepared.
US-30 AI ID: A1	Warehouse Employee	User Story: As a warehouse employee, I want to mark a completed warehouse order as ready for shipping so that the shipping process can continue.	AT-30A: GIVEN all required warehouse work for an order has been completed, WHEN the employee marks the order ready for shipping, THEN the system records the order as ready for shipping. AT-30B: GIVEN required products have not all been successfully prepared, WHEN the employee attempts to mark the order ready for shipping, THEN the order is not recorded as ready for shipping. AT-30C: GIVEN an order has been marked ready for shipping, WHEN the order is viewed afterward, THEN its status reflects that it is ready for shipping.

*Paste the AI draft verbatim. Do not improve wording here. Use the AI evidence ID that produced each item.****Need more rows? Add rows while keeping the same fields. Your submitted backlog should remain approximately 4-6 stories per role.***

WORKSHEET 2

Revision log

Record changes from human review, prioritization, dependency analysis, and prototype walkthroughs. Do not use AI for these decisions.

STEP	ID	PROBLEM / EVIDENCE	CHANGE	RATIONALE
Step 2	US-02	What statement says, "I want to order available sporting goods so that I can purchase products from CUSports." Ordering and purchasing is the same thing.	Will replace with, "I want to order available sporting goods so that I don't have to go to a physical store."	Gets rid of the repetitive story while keeping the need the same.
Step 2	US-06	No test states what happens if someone without authorization tries to operate the CUSports System	Will add test AT-06e, "Given a non-administrator is authenticated, when they attempt any user management action, then the system denies the action and makes no change."	Improves the security of the software, adds story specified authorized user test for statement, "authorized users can operate the CUSports system."
Step 2	US-17	Story combines preparing an order for shipment with verification of order correctness ("so that I can process the correct order and send it to the correct destination").	Replace with "I want to view the information needed to prepare an order for shipment, so that I can process the order with the given shipping information."	This narrows the goal to one outcome.
Step 2	US-21	The "current status" of a customer's order is a bit vague. The acceptance tests reflect that the actual tracking of orders according to a specific customer is the goal.	As a customer service representative, I want to be able to view orders purchased by a customer, so I can locate and answer questions about the customer's order	This improves the requirement because it Aligns with the generated acceptance tests, and narrows down the scope from "status" to "purchase tracking".
Step 6	AT-02B	Currently unavailable items can be added to orders only to be later removed	Unavailable items cannot be added to orders	Reduces confusion by not allowing for unavailable items to be added to orders just to later be removed.
Step 3	US-29	Misleading in it's effectiveness and practicality in real world scenarios. Adds unnecessary time and costs.	Removed from the user story list.	This should be a back end feature or one time input (by human if needed) that is then input into a database, that then reflects the current storage of a certain item or order given a database.
[Step 2/3/4/6]	[US/AT ID]	[Problem or evidence]	[Change made]	[Why this improves the requirement]
[Step 2/3/4/6]	[US/AT ID]	[Problem or evidence]	[Change made]	[Why this improves the requirement]
[Step 2/3/4/6]	[US/AT ID]	[Problem or evidence]	[Change made]	[Why this improves the requirement]

Examples of Step evidence: quality-check failure, inability to prioritize, hidden prerequisite, circular dependency, confusing prototype interaction, or an acceptance test that could not be demonstrated.

WORKSHEET 3

Prioritized backlog

Assign every story a MoSCoW category and a unique rank. Mark 3–5 stories that form one coherent MVP workflow.

3-5 stories marked with pink highlight

RANK	ID	MoSCoW	VALUE	RISK / LEARNING	RATIONALE
1	[US-01]	[M/S/C/W]	[H/M/L]	[H/M/L]	It's the basis for the app. Without being able to see the products, there is nothing to build off of.
2	[US-02]	[M/S/C/W]	[H/M/L]	[H/M/L]	This is the core revenue of the app, but it is reliant on US-01
3	[US-08]	[M/S/C/W]	[H/M/L]	[H/M/L]	Without a way to set availability, US-01 and US-02 will have nothing accurate to work from. Put behind them because they still work without this, just not accurately.
4	[US-03]	[M/S/C/W]	[H/M/L]	[H/M/L]	Needed so customers trust what they are buying, or else they won't be buying at all.
5	[US-12]	[M/S/C/W]	[H/M/L]	[H/M/L]	Essential for ensuring stock is maintained for the customers.
6	[US-16]	[M/S/C/W]	[H/M/L]	[H/M/L]	A shipper must be able to recognize which orders are available to be processed in order to actually process them.
7	[US-17]	[M/S/C/W]	[H/M/L]	[H/M/L]	Without being able to view relevant shipping information, a shipper cannot process an order.
8	[US-20]	[M/S/C/W]	[H/M/L]	[H/M/L]	Without the assumption that shipping is handled in-house, multiple different shippers could potentially be interacting with the system. With this in mind, they need to be able to see whether a shipment is available to be processed by them or not.
9	[US-21]	[M/S/C/W]	[H/M/L]	[H/M/L]	The user needs to see if their order has gone through, in case of any issues on their end.
10	[US-22]	[M/S/C/W]	[H/M/L]	[H/M/L]	The user needs to see details of the items in their order, and if their inquiries can be solved via CSR.
11	[US-26]	[M/S/C/W]	[H/M/L]	[H/M/L]	The warehouse worker needs to be able to see the required orders to prepare. Without knowing what orders to prepare, then CUSports would not have product moving out of the warehouse, ruining the point of the app.
12	[US-30]	[M/S/C/W]	[H/M/L]	[H/M/L]	Making sure that orders are marked complete will allow for shipping to carry out its role and is essential for ensuring timely deliveries.
13	[US-09]	[M/S/C/W]	[H/M/L]	[H/M/L]	It's needed to move orders towards fulfillment, but it's an internal workflow, not customer facing.
14	[US-04]	[M/S/C/W]	[H/M/L]	[H/M/L]	Reduces support inquiries, but depends on US-09 having shipping data to show.
15	[US-07]	[M/S/C/W]	[H/M/L]	[H/M/L]	Basic catalogue maintenance work that can happen in parallel without blocking the other user stories.
16	[US-06]	[M/S/C/W]	[H/M/L]	[H/M/L]	Needed for internal operations, but has no direct customer value and can be handled manually early release.
17	[US-11]	[M/S/C/W]	[H/M/L]	[H/M/L]	Useful for showing profits over a certain time period and for running the business, but not essential for the store to function.
18	[US-15]	[M/S/C/W]	[H/M/L]	[H/M/L]	Useful for identifying improvements in the store, or fixing issues/bugs that customers discover

RANK	ID	MoSCoW	VALUE	RISK / LEARNING	RATIONALE
19	[US-18]	[M/S/C/W]	[H/M/L]	[H/M/L]	While not necessary to complete the function of a shipper, having receipts/confirmation of delivery is industry standard and important for record keeping.
20	[US-23]	[M/S/C/W]	[H/M/L]	[H/M/L]	The customer SHOULD be able to see product availability, and CSR interaction should be limited to more specialized cases, but in case of undefined behavior on the customer's side, it should be a thing.
21	[US-24]	[M/S/C/W]	[H/M/L]	[H/M/L]	A customer's view of their status may not be super intuitive, but if it is, CSR may not help a lot.
22	[US-25]	[M/S/C/W]	[H/M/L]	[H/M/L]	It is important to look up the user's order given a certain, unique id, but may be overengineering given US-21, but should be painless to implement.
23	[US-05]	[M/S/C/W]	[H/M/L]	[H/M/L]	Improves retention, but alternatives such as emails can be used until this is built.
24	[US-10]	[M/S/C/W]	[H/M/L]	[H/M/L]	No customer impact. Isn't foundational to the needs of the consumers and will probably be the least used function.
25	[US-13]	[M/S/C/W]	[H/M/L]	[H/M/L]	Not essential for the store to run, but helpful for displaying data to shareholders or other company representatives.
26	[US-14]	[M/S/C/W]	[H/M/L]	[H/M/L]	Not essential for the store to run, but helpful for displaying high value items to shareholders or other company representatives.
27	[US-19]	[M/S/C/W]	[H/M/L]	[H/M/L]	This is an unnecessary function for shippers as they do not need to know what shipments aren't ready for processing, although it could be nice to know if there are shipments in the process of being prepared but are not ready yet.
28	[US-27]	[M/S/C/W]	[H/M/L]	[H/M/L]	Potentially overengineering, if this requirement filters warehousing packaging/processing per specific order, may induce granular prep that a sequential order #ID and the corresponding products under that can follow.
29	[US-28]	[M/S/C/W]	[H/M/L]	[H/M/L]	Low immediate need. Involved engineering for non standard edge cases and adds moderate risks that can break.
30	[US-29]	[M/S/C/W]	[H/M/L]	[H/M/L]	Sounds good at first glance, but this is likely something that should not be handled by warehouse workers. This should be a back end feature that reflects the current storage of a certain item or order given a database.

Add rows for the additional stories while keeping the same fields.

MoSCoW Categories:

1. **Must-Have:** Non-negotiable product needs that are mandatory for the project/product
2. **Should Have:** Important stories that are not vital, but add significant value
3. **Could Have:** Nice-to-have stories that will have a small impact if left out
4. **Will Not Have:** Stories that are not priority for this specific time frame

WORKSHEET 4

Dependency identification and response

List true prerequisites in the correct direction. A related story is not automatically a dependency.

DEPENDENT STORY	PREREQUISITE	TYPE	JUSTIFICATION	SEQUENCING / RESPONSE
[US-01]	[US-__ / decision / system]	Data	There's no actual warehouse to give real data	Mock
[US-02]	[US-01 / decision / system]	Data	US-01 is required first so that a customer knows if the item they are ordering is actually in stock	Defer
[US-03]	[US-02 / decision / system]	Workflow order	US-02 is required first so that a customer can have an active order that has information that can be viewed	Defer
[US-04]	[US-18 / decision / system]	External	The shipping department needs to provide this information before the customer can view it	Defer
[US-05]	[US-__ / decision / system]	External	There needs to be a customer service department to contact	Mock
[US-06]	[US-__ / decision / system]	Capability	Business Admins must have access to user-management functionality	Defer
[US-07]	[US-__ / decision / system]	Data	There's no products or product data to provide as of yet.	Mock
[US-08]	[US-__ / decision / system]	Data	There's no actual warehouse to give real data	Mock
[US-09]	[US-__ / decision / system]	Data	No warehouse or shipping department to provide information	Mock
[US-10]	[US-__ / decision / system]	Data	No purchases or operations have been made	Mock
[US-11]	[US-10 / decision / system]	Capability	A sales report falls under the business reports that US-10 would create.	Defer
[US-12]	[US-29 / decision / system]	External	The warehouse must update the stock of items in the system for an accurate report of system stock to be generated	Sequence
[US-13]	[US-11 / decision / system]	Workflow Order	US-11 generates the report that the user will be able to download.	Split
[US-14]	[US-11 / decision / system]	Workflow order	US-11 generates a sales report over a date range, and this data can be inferred from that.	Split
[US-15]	[US-05 / decision / system]	Capability	US-05 specifies that there needs to be customer support. Without customer support, there wouldn't be any support tickets.	Defer
[US-16]	[US-09 / decision / system]	Shared Component	Without a system that stores a state determining shipment readiness status, the information cannot be displayed to the shipper.	Defer

DEPENDENT STORY	PREREQUISITE	TYPE	JUSTIFICATION	SEQUENCING / RESPONSE
[US-17]	[US-09 / decision / system]	Shared Component	Without a system that stores information related to shipment information like destination, appointment times, etc., the information cannot be displayed to the shipper.	Defer
[US-18]	[US-__ / decision / system]	Capability	There needs to be a way for the shipper to interact with the system to update the state of the shipment.	Sequence
[US-19]	[US-09 / decision / system]	Shared Component	Without a system that stores a state determining shipment readiness status and corresponding reasons, that information cannot be displayed to the shipper.	Defer
[US-20]	[US-09 / decision / system]	Shared Component	Without a system that stores a state determining shipment readiness status, the information cannot be displayed to the shipper.	Defer
[US-21]	US-23, US-25, US-24	Capability or Data	“Status is ambiguous”, but in this case, shipping status and availability status can fall under that.	Split
[US-22]	US-25	Capability or data	The data (specific customer order identifier logic) is needed to view products within a customer’s order.	Sequence
[US-23]	US-08	Capability or Data	CSR would likely be reading from the same data source customers do.	Sequence
[US-24]	US-25	Capability or Data	CSR has to locate the order before viewing its shipping status, US-25 is all about order IDs.	Sequence
[US-25]	US-06	Policy or Decision	CSR must get authorization access to that data.	Sequence
[US-26]	US-02	Capability or Data	An employee would select an order before viewing its required items	Sequence
[US-27]	US-26, US-02	Workflow Order	The customer fills out an order, then it is reflected to the warehouse employee.	Sequence
[US-28]	US-27	Workflow Order	One needs the required items list before marking an item as picked	Sequence
[US-29]	US - 27	Workflow order	One needs the required items list and their availability before reporting unavailability.	Sequence
[US-30]	US-28	Workflow order	AT-30a also needs “ready for shipping” to confirm picks being complete	Sequence

Add rows for the dependencies while keeping the same fields.

WORKSHEET 5

Human-authored prototype plan

Complete this plan before asking AI to generate prototype code. You may want to break up the prototype into different screens based on the role.

PLAN ITEM	TEAM DECISION
Primary user	Customer - Wants to use the software to buy Clemson sports products without having to visit the store in person.
End-to-end task	Browse and purchase items on the CUSports website
Story IDs	US-01, US-02, US-03, US-04
Acceptance-test IDs to demonstrate	AT-01a, AT-01B, At-01C, AT-01D, AT-02a, AT-02B, AT-02C, AT-02D, AT-03a, AT-03B, AT-03C, AT-03D, AT-04a, AT-04B, AT-04C, AT-04D
Pages / screens / components	Home screen, product screen, payment screen, order detail screen
Normal-path interactions	User can click items to view them, click on a search bar, choose to purchase an item, and submit payment
Alternate / validation / error state	Validate that the user's payment goes through
Synthetic data needed	Clemson sports products, merch, customers
Out of scope	All user stories that do not fall under the "Must have" category is MoSCoW
Technical constraints for the AI prompt	[Local index.html, files, accessibility, no external services, etc.] No external services, start file: index.html, semantic HTML, keyboard-operable controls, synthetic data.

Prototype file inventory

Start file: index.html Supporting files: [app.js](#), [data.js](#), styleless.css, README.txt

AI evidence IDs used for prototype creation: A3

Include any updated prompts needed after testing.

WORKSHEET 6

Manual acceptance-test walkthrough

Run the mapped tests against the prototype. Use: Demonstrated, Partially demonstrated, Not represented, or Blocked for your Result Column.

TEST	SETUP / ACTION	OBSERVED EVIDENCE	RESULT	DECISION / CHANGE
[AT-01a]	Given a product with availability information, when the customer views the product, then the system displays the product's current availability status.	A green box that says available	Demonstrated	No change because the test is cleared, and the result is demonstrated.
[AT-01B]	Given a product that is not available, when the customer views the product, then the system does not indicate that the product is available.	A red box that says out of stock	Demonstrated	No change because the test is cleared, and the result is demonstrated.
[AT-01C]	Given a product whose availability information cannot be determined, when the customer views the product, then the system does not incorrectly indicate that the product is available.	A yellow box that says availability unknown	Demonstrated	No change because the test is cleared, and the result is demonstrated.
[AT-01D]	Given a customer views a product, when the availability status is displayed, then the status corresponds to the availability information maintained by the system.	A box that correlates to the availability with the corresponding color	Demonstrated	No change because the mock system data correlates with represented availability.
[AT-02a]	Given an order containing available product(s), when the customer submits the order, then the system creates the order.	Within "My Orders" tab, purchased items show up as orders	Demonstrated	No change because the system recognizes that a completed purchase becomes an order and organizes it into a specific tab
[AT-02B]	Given an order containing a product that is unavailable, when the customer submits the order, then the system does not create an order that includes the unavailable product.	There's no evidence of this happening because unavailable items cannot be added to orders to begin with	Not represented	Reword test to reduce confusion of being able to add unavailable items to orders only for them to be removed upon submission.
[AT-02C]	Given an order containing multiple products, when one required product cannot be ordered, then the system does not silently create an order containing a different set of products.	Order containing unavailable product and an available product cannot be created into an order	Demonstrated	No change because an order cannot be created with an unavailable product
[AT-02D]	Given a successfully submitted order, when the order is created, then the system associates the ordered product(s) with that order.	Orders are created as groups under the "My Orders" tab	Demonstrated	No change because groups of items in one order and grouped together into one order under the "My Orders" tab.
[AT-03a]	Given a customer has an order, when the customer accesses that order, then the system displays information identifying the order.	Order tracking number, cost, carrier, delivery status/date are all present	Demonstrated	No change because all order information is given with each order
[AT-03B]	Given an order contains one or more products, when the customer views the order, then the system displays the products associated with that order.	Order containing several items displays each item ordered and in what quantity	Demonstrated	No change because the items and their quantity are shown in the orders.

TEST	SETUP / ACTION	OBSERVED EVIDENCE	RESULT	DECISION / CHANGE
[AT-03C]	Given a customer has no orders, when the customer accesses their orders, then the system does not display another customer's order as their own.	The "My Orders" tab shows that there are no orders for a different user.	Demonstrated	No change because users have different "My Orders" sections that allow for personalized and private viewings of their own orders
[AT-03D]	Given an order exists, when the customer views it, then the displayed order information reflects the information stored for that order.	The information stored for the order reflects the information at the bottom of the order	Demonstrated	No change because the system information reflects the information given to the customer
[AT-04a]	Given an order has shipping information, when the customer views the order, then the system displays the available shipping information.	Shipping information is included at the bottom of each order	Demonstrated	No change because the shipping information is provided with each order including shipping company and ID.
[AT-04B]	Given an order does not yet have shipping information, when the customer views the order, then the system does not display fabricated shipping information.	Orders that have not yet been shipped do not contain any shipping information	Demonstrated	No change because shipping information is not provided to un-shipped orders.
[AT-04C]	Given shipping information changes, when the customer views the order after the change has been recorded, then the displayed information reflects the recorded information.	Upon simulating a shipping update, shipping information appears for the order	Demonstrated	No change because shipping information is added when simulating a shipping change
[AT-04D]	Given a customer has multiple orders, when the customer views shipping information for one order, then the system displays shipping information associated with that order rather than another order.	Shipping information is different between each order	Demonstrated	No change because shipping information is unique between orders.

Add rows for the additional tests while keeping the same fields.

WORKSHEET 7

Final story and acceptance-test records

Use one record per final story; add records as needed. Preserve IDs and reference all dependencies and prototype coverage.

FINAL STORY 1 · US-01 · ROLE: Customer · RANK: 1 · MOSCOW: M · MVP: Y
Story

As a customer, I want to view whether a sporting goods product is available so that I can determine whether it can be ordered.

Acceptance tests

AT-01a: Given a product with availability information, when the customer views the product, then the system displays the product's current availability status.

AT-01B: Given a product that is not available, when the customer views the product, then the system does not indicate that the product is available.

AT-01C: Given a product whose availability information cannot be determined, when the customer views the product, then the system does not incorrectly indicate that the product is available.

AT-01D: Given a customer views a product, when the availability status is displayed, then the status corresponds to the availability information maintained by the system.

Dependencies / sequence: System Prototype coverage: AT-01a, AT-01B, AT-01C, AT-01D

FINAL STORY 2 · US-02 · ROLE: Customer · RANK: 2 · MOSCOW: M · MVP: Y
Story

As a customer, I want to order available sporting goods so that I don't have to go to a physical store.

Acceptance tests

AT-02a: Given an order containing available product(s), when the customer submits the order, then the system creates the order.

AT-02B: Given an order containing a product that is unavailable, when the customer submits the order, then the system does not create an order that includes the unavailable product.

AT-02C: Given an order containing multiple products, when one required product cannot be ordered, then the system does not silently create an order containing a different set of products.

AT-02D: Given a successfully submitted order, when the order is created, then the system associates the ordered product(s) with that order.

Dependencies / sequence: US-01 Prototype coverage: AT-02a, AT-02B, AT-02C, AT-02D

FINAL STORY 3 · US-03 · ROLE: Customer · RANK: 4 · MOSCOW: M · MVP: Y
Story

As a customer, I want to view information about my order so that I can understand what I ordered..

Acceptance tests

AT-03a: Given a customer has an order, when the customer accesses that order, then the system displays information identifying the order.

AT-03B: Given an order contains one or more products, when the customer views the order, then the system displays the products associated with that order.

AT-03C: Given a customer has no orders, when the customer accesses their orders, then the system does not display another customer's order as their own.

AT-03D: Given an order exists, when the customer views it, then the displayed order information reflects the information stored for that order.

Dependencies / sequence: US-02 Prototype coverage: AT-03a, AT-03B, AT-03C, AT-03D

FINAL STORY 4 · US-04 · ROLE: Customer · RANK: 14 · MOSCOW: S · MVP: N

Story

As a customer, I want to access shipping information for my order so that I can understand the status of its shipment.

Acceptance tests

AT-04a: Given an order has shipping information, when the customer views the order, then the system displays the available shipping information.

AT-04B: Given an order does not yet have shipping information, when the customer views the order, then the system does not display fabricated shipping information.

AT-04C: Given shipping information changes, when the customer views the order after the change has been recorded, then the displayed information reflects the recorded information.

AT-04D: Given a customer has multiple orders, when the customer views shipping information for one order, then the system displays shipping information associated with that order rather than another order.

Dependencies / sequence: US-18 Prototype coverage: AT-04a, AT-04B, AT-04C, AT-04D

FINAL STORY 5 · US-05 · ROLE: Customer · RANK: 23 · MOSCOW: C · MVP: N

Story

As a customer, I want to contact CUSports customer support so that I can request assistance with my order or account.

Acceptance tests

AT-05a: Given a customer has access to the customer-support mechanism, when the customer submits a support request, then the system records the submitted request.

AT-05B: Given a customer submits a support request, when required information is missing, then the system does not record the request as complete.

AT-05C: Given a customer submits a support request, when the submission succeeds, then the system provides an observable indication that the request was submitted.

AT-05D: Given a customer submits a support request concerning an order, when the request is recorded, then the order information supplied by the customer remains associated with the request.

Dependencies / sequence: System Prototype coverage: AT-05a, AT-05B, AT-05C, AT-05D

FINAL STORY 6 · US-06 · ROLE: System Admin · RANK: 16 · MOSCOW: S · MVP: N

Story

As a system administrator, I want to manage system-user access so that authorized users can operate the CUSports system.

Acceptance tests

AT-06a: Given the administrator has access to user-management functionality, when the administrator views system users, then the system displays the users available for management.

AT-06B: Given the administrator performs a supported user-management action, when the action succeeds, then the resulting user information reflects that action.

AT-06C: Given the administrator attempts to perform an unsupported or invalid user-management action, then the system does not apply the invalid change.

AT-06D: Given a user-management change is successfully made, when the administrator views the affected user afterward, then the displayed information reflects the saved change.

AT-06E: Given a non-administrator is authenticated, when they attempt any user management action, then the system denies the action and makes no change.

Dependencies / sequence: System Prototype coverage: Not Represented

FINAL STORY 7 · US-07 · ROLE: System Admin · RANK: 15 · MOSCOW: S · MVP: N

Story

As a system administrator, I want to maintain product information so that CUSports can provide accurate product data.

Acceptance tests

AT-07a: Given an administrator has access to product management, when the administrator views a product, then the system displays the product information stored for that product.

AT-07B: Given the administrator makes a supported change to product information, when the change is saved successfully, then the system displays the updated information.

AT-07C: Given the administrator attempts to save invalid or incomplete product information where validation is required, then the system does not save the invalid information.

AT-07D: Given an administrator changes product information, when a customer subsequently views that product, then the customer sees the product information currently maintained by the system.

Dependencies / sequence: System Prototype coverage: Not Represented

FINAL STORY 8 · US-08 · ROLE: System Admin · RANK: 3 · MOSCOW: M · MVP: Y

Story

As a system administrator, I want to maintain product availability so that customers receive accurate availability information and can order products that are available.

Acceptance tests

AT-08a: Given an administrator has access to availability management, when the administrator records a supported availability change, then the system saves the new availability information.

AT-08B: Given a product is marked unavailable, when a customer attempts to order that product, then the system does not treat the product as available.

AT-08C: Given a product's availability is changed from unavailable to available, when the change is successfully saved, then the system uses the updated availability information.

AT-08D: Given an administrator attempts to save an invalid availability value, then the system does not save the invalid value.

Dependencies / sequence: System Prototype coverage: Not Represented

FINAL STORY 9 · US-09 · ROLE: System Admin · RANK: 13 · MOSCOW: S · MVP: N

Story

As a system administrator, I want to maintain the system information needed for warehouse and shipping operations so that orders can be processed and shipped accurately.

Acceptance tests

AT-09a: Given an order exists in the system, when an authorized warehouse or shipping operation updates supported order information, then the system records the update.

AT-09B: Given an order does not exist, when an operation attempts to update that order, then the system does not create or update an unrelated order.

AT-09C: Given required information for a supported warehouse or shipping update is missing, when the update is submitted, then the system does not save the incomplete update.

AT-09D: Given a warehouse or shipping update is successfully recorded, when the order is subsequently viewed, then the system displays the recorded information.

Dependencies / sequence: System Prototype coverage: Not Represented

FINAL STORY 10 · US-10 · ROLE: System Admin · RANK: 24 · MOSCOW: C · MVP: N

Story

As a system administrator, I want to generate business reports so that I can review information about CUSports operations.

Acceptance tests

AT-10a: Given the administrator has access to a supported report, when the administrator requests the report, then the system generates the report using the available system data.

AT-10B: Given the requested report has no matching data, when the administrator generates the report, then the system indicates that no matching data is available rather than displaying unrelated data.

AT-10C: Given a report-generation request contains an invalid or unsupported parameter, then the system does not generate a report using the invalid parameter as though it were valid.

AT-10D: Given a report is successfully generated, when the administrator views it, then the report reflects the underlying system data used to generate it.

Dependencies / sequence: System Prototype coverage: Not Represented

FINAL STORY 11 · US-11 · ROLE: Business User · RANK: 17 · MOSCOW: S · MVP: N

Story

As a Business User, I want to view a sales report for a date range I select, so that I can understand revenue and order volume performance over a specific period.

Acceptance tests

AT-11a: Given a business user is logged in and on the reporting section, when they select a valid start date and end date and submit, then a report displays total number of orders and total sales revenue for that exact range.

AT-11B: Given a business user enters a start date that is after the end date, when they submit, then the system displays a validation error and no report is generated.

AT-11C: Given a business user selects a date range with zero completed orders, when they submit, then the report displays zero orders and \$0.00 revenue rather than an error or blank screen.

AT-11D: Given a business user selects a date range spanning more than one calendar year, when they submit, then the report still returns accurate totals for the full selected range (no silent truncation).

Dependencies / sequence: US-10, Defer Prototype coverage: Not Represented

FINAL STORY 12 · US-12 · ROLE: Business User · RANK: 5 · MOSCOW: M · MVP: N

Story

As a Business User, I want to view a report of products at or below a defined stock threshold, so that I can identify items that may need reordering.

Acceptance tests

AT-12a: Given products exist with stock quantities below the defined threshold, when the business user opens the low-stock report, then all such products are listed with their product name and current quantity.

AT-12B: Given no products are currently below the threshold, when the business user opens the report, then the system displays a clear "no low-stock items" message rather than an empty/blank table.

AT-12C: Given a product's stock quantity changes after the report is generated, when the business user refreshes/regenerates the report, then the updated quantity is reflected.

AT-12D: Given a product has a stock quantity of exactly zero, when the business user views the report, then that product appears in the report (zero counts as below/at threshold).

Dependencies / sequence: US-08, Sequence Prototype coverage: Not Represented

FINAL STORY 13 · US-13 · ROLE: Business User · RANK: 25 · MOSCOW: C · MVP: N

Story

As a Business User, I want to export a generated report to a downloadable file, so that I can share or analyze the data outside the system.

Acceptance tests

AT-13a: Given a report is displayed on screen, when the business user selects "Export," then a file downloads containing the same data shown on screen.

AT-13B: Given a report contains zero rows of data, when the business user exports it, then the downloaded file is still generated successfully with headers only (no error/crash).

AT-13C: Given an export is in progress, when the business user attempts to trigger a second export before the first completes, then the system either queues, blocks, or clearly indicates the export is already running (no duplicate/corrupted files).

AT-13D: Given the exported file is opened, when the business user inspects it, then column headers match the on-screen report labels exactly.

Dependencies / sequence: US-11, Split Prototype coverage: Not Represented

FINAL STORY 14 · US-14 · ROLE: Business User · RANK: 26 · MOSCOW: C · MVP: N

Story

As a Business User, I want to view a ranked list of top-selling products over a date range, so that I can identify which products are driving the most business.

Acceptance tests

AT-14a: Given completed orders exist within a selected date range, when the business user requests the top-selling products report, then products are listed in descending order by units sold.

AT-14B: Given two or more products have the exact same units sold, when the report is generated, then both products appear in the report without one being dropped or overwritten.

AT-14C: Given a selected date range has no completed orders, when the report is generated, then the system displays a clear empty-state message instead of an error.

AT-14D: Given a product was sold during the range but has since been removed/discontinued from the catalog, when the report is generated, then it still appears in historical results (report reflects sales history, not current catalog status).

Dependencies / sequence: US-11, Split Prototype coverage: Not Represented

FINAL STORY 15 · US-15 · ROLE: Business User · RANK: 18 · MOSCOW: S · MVP: N

Story

As a Business User, I want to view a report showing the number of customer support tickets opened and resolved over a date range, so that I can monitor customer support workload and performance trends.

Acceptance tests

AT-15a: Given tickets were opened and/or resolved within a selected date range, when the business user generates the report, then it shows separate counts for tickets opened and tickets resolved in that range.

AT-15B: Given a ticket was opened before the selected range but resolved within it, when the report is generated, then it is correctly counted only in the "resolved" total for that range, not "opened."

AT-15C: Given no tickets were opened or resolved in the selected range, when the report is generated, then it displays zero counts rather than an error or blank output.

AT-15D: Given a ticket is still open/unresolved as of the report date, when the report is generated, then it is excluded from the "resolved" count but included in the "opened" count if applicable to the range.

Dependencies / sequence: US-05, Defer Prototype coverage: Not Represented

FINAL STORY 16 · US-16 · ROLE: Shipper · RANK: 6 · MOSCOW: M · MVP: N

Story

As a Shipper, I want to view orders that are ready for shipment, so that I can identify which orders need to be processed.

Acceptance tests

AT-16a: Given an order meets the system's existing criteria for being ready to ship, when the shipper views the ready-to-ship orders, then that order is displayed.

AT-16B: Given an order does not meet the criteria for being ready to ship, when the shipper views the ready-to-ship orders, then that order is not displayed.

AT-16C: Given there are no orders currently ready to ship, when the shipper views the list, then the system displays an observable indication that there are no ready-to-ship orders.

AT-16D: Given multiple orders are ready to ship, when the shipper views the list, then each qualifying order is displayed as a distinct order that the shipper can identify.

Dependencies / sequence: US-09, Defer Prototype coverage: Not Represented

FINAL STORY 17 · US-17 · ROLE: Shipper · RANK: 7 · MOSCOW: M · MVP: Y

Story

As a Shipper, I want to view the information needed to prepare an order for shipment, so that I can process the order with the given shipping information..

Acceptance tests

AT-17a: Given the shipper selects an order available for shipment, when the order details are opened, then the system displays the shipment information defined as required for that order.

AT-17B: Given the selected order contains multiple products, when the shipper opens its details, then each product included in the order is displayed.

AT-17C: Given an order has incomplete required shipment information, when the shipper opens the order, then the system identifies the missing information rather than presenting the order as having complete shipment information.

AT-17D: Given the shipper is viewing one order, when the shipper opens a different order, then the displayed shipment information corresponds to the newly selected order.

Dependencies / sequence: US-09, Defer Prototype coverage: Not Represented

FINAL STORY 18 · US-18 · ROLE: Shipper · RANK: 19 · MOSCOW: S · MVP: N

Story

As a Shipper, I want to record that an order has been shipped, so that CUSports has an up-to-date record of the order's shipping status.

Acceptance tests

AT-18a: Given an order is eligible to be marked as shipped, when the shipper records the shipment as complete, then the order's status changes to the system's defined shipped status.

AT-18B: Given an order has already been marked as shipped, when the shipper attempts to mark it as shipped again, then the system does not create a second shipment completion for the same order.

AT-18C: Given an order is not eligible to be marked as shipped, when the shipper attempts to record shipment completion, then the system does not mark the order as shipped.

AT-18D: Given the shipment is successfully recorded, when the shipper returns to the order, then the order displays its updated shipped status.

Dependencies / sequence: System, Sequence Prototype coverage: Not Represented

FINAL STORY 19 · US-19 · ROLE: Shipper · RANK: 27 · MOSCOW: C · MVP: N

Story

As a Shipper, I want to identify orders that cannot currently be shipped, so that I can avoid incorrectly completing their shipment.

Acceptance tests

AT-19a: Given an order has a system-recognized shipment-blocking condition, when the shipper views the order, then the system indicates that the order cannot currently be shipped.

AT-19B: Given an order has no system-recognized shipment-blocking condition, when the shipper views the order, then the system does not identify it as having a shipment exception.

AT-19C: Given an order has a shipment exception, when the shipper attempts to complete shipment, then the system prevents the shipment from being recorded as completed.

AT-19D: Given the shipment-blocking condition is removed, when the shipper subsequently views the order, then the order no longer shows that exception if the system's rules determine it is now eligible for shipment.

Dependencies / sequence: US-09, Defer Prototype coverage: Not Represented

FINAL STORY 20 · US-20 · ROLE: Shipper · RANK: 8 · MOSCOW: M · MVP: N

Story

As a Shipper, I want to view the current shipping status of an order, so that I can determine whether the order still requires shipping work.

Acceptance tests

AT-20a: Given an order has a recorded shipping status, when the shipper views the order, then the current status is displayed.

AT-20b: Given an order's shipping status has been updated, when the shipper views the order after the update, then the newly recorded status is displayed.

AT-20c: Given an order has no valid shipping status, when the shipper views the order, then the system does not display an incorrect status and instead indicates that the status is unavailable or invalid.

AT-20d: Given two orders have different shipping statuses, when the shipper views each order, then each order displays its own corresponding status.

Dependencies / sequence: US-09, Defer Prototype coverage: Not Represented

FINAL STORY 21 · US-21 · ROLE: Customer Service Representative · RANK: 9 · MOSCOW: M · MVP: N

Story

As a customer service representative, I want to be able to view orders purchased by a customer, so I can locate and answer questions about the customer's order.

Acceptance tests

AT-21a: GIVEN a valid order exists, WHEN the CSR opens the order, THEN the system displays the order's current status.

AT-21b: GIVEN an order cannot be found, WHEN the CSR searches for it, THEN the system indicates that no matching order was found.

AT-21c: GIVEN an order's status changes, WHEN the CSR views the order afterward, THEN the displayed status reflects the updated status.

Dependencies / sequence: US-23,24,25 Prototype coverage: Not Represented

FINAL STORY 22 · US-22 · ROLE: Customer Service Representative · RANK: 10 · MOSCOW: M · MVP: N

Story

As a customer service representative, I want to view the products included in a customer's order so that I can answer questions about what the customer purchased.

Acceptance tests

AT-22a: GIVEN a valid order exists, WHEN the CSR views its details, THEN the system displays the products associated with the order.

AT-22b: GIVEN an order contains multiple products, WHEN the CSR views the order, THEN each ordered product is shown separately.

AT-22c: GIVEN an order has a recorded quantity for a product, WHEN the CSR views that product in the order, THEN the correct ordered quantity is displayed.

Dependencies / sequence US-25 Prototype coverage: Not Represented

FINAL STORY 23 · US-23 · ROLE: Customer Service Representative · RANK: 20 · MOSCOW: S · MVP: N

Story

As a customer service representative, I want to check whether a product is currently available so that I can provide customers with accurate availability information.

Acceptance tests

AT-23a: GIVEN an available product, WHEN the CSR checks its availability, THEN the system reports that the product is available.

AT-23b: GIVEN an unavailable product, WHEN the CSR checks its availability, THEN the system reports that the product is unavailable.

AT-23c: GIVEN product availability changes, WHEN the CSR checks the product afterward, THEN the current availability is displayed.

Dependencies / sequence: US-08 Prototype coverage: Not Represented

FINAL STORY 24 · US-24 · ROLE: Customer Service Representative · RANK: 21 · MOSCOW: C · MVP: N

Story

As a customer service representative, I want to view the shipping status of an order so that I can answer customer questions about its shipment.

Acceptance tests

AT-24a: GIVEN an order has entered the shipping process, WHEN the CSR views the order, THEN the system displays its current shipping status.

AT-24B: GIVEN an order has not entered the shipping process, WHEN the CSR views shipping information, THEN the system indicates that shipping information is not yet available.

AT-24C: GIVEN the shipping status has changed, WHEN the CSR views the order afterward, THEN the updated shipping status is displayed.

Dependencies / sequence: US-25 Prototype coverage: Not Represented

FINAL STORY 25 · US-25 · ROLE: Customer Service Representative · RANK: 22 · MOSCOW: C · MVP: N

Story

As a customer service representative, I want to locate a customer order using its order identifier so that I can access the correct order while assisting the customer.

Acceptance tests

AT-25a: GIVEN a valid order identifier, WHEN the CSR searches for it, THEN the matching order is displayed.

AT-25B: GIVEN an identifier that does not match an order, WHEN the CSR searches for it, THEN the system reports that no matching order was found.

AT-25C: GIVEN multiple orders exist in the system, WHEN the CSR searches using a specific valid order identifier, THEN only the matching order is returned.

Dependencies / sequence: US-06, US-02 Prototype coverage: Not Represented

FINAL STORY 26 · US-26 · ROLE: Warehouse Employee · RANK: 11 · MOSCOW: S · MVP: N

Story

As a warehouse employee, I want to view orders that require warehouse work so that I know which orders need to be prepared.

Acceptance tests

AT-26a: GIVEN orders are awaiting warehouse work, WHEN the warehouse employee views the work list, THEN those orders are displayed.

AT-26B: GIVEN an order no longer requires warehouse work, WHEN the work list is refreshed, THEN that order is no longer shown as awaiting work.

AT-26C: GIVEN no orders currently require warehouse work, WHEN the employee views the work list, THEN the system indicates that no orders are awaiting warehouse work.

Dependencies / sequence: US-02 Prototype coverage: Not Represented

FINAL STORY 27 · US-27 · ROLE: Warehouse Employee · RANK: ____ · MOSCOW: ____ · MVP: Y / N

Story

As a warehouse employee, I want to view the products and quantities required for an order so that I can prepare the correct items.

Acceptance tests

AT-27a: GIVEN a valid warehouse order, WHEN the employee opens it, THEN the required products are displayed.

AT-27B: GIVEN a product has a required quantity greater than one, WHEN the employee views the order, THEN the required quantity is displayed.

AT-27C: GIVEN an order contains multiple products, WHEN the employee opens it, THEN all products associated with that order are shown.

Dependencies / sequence: US-26, US-02 Prototype coverage: Not Represented

FINAL STORY 28 · US-28 · ROLE: Warehouse Employee · RANK: 29 · MOSCOW: C · MVP: N

Story

As a warehouse employee, I want to record when required products have been picked so that the system reflects the progress of warehouse work.

Acceptance tests

AT-28a: GIVEN a required product has not been picked, WHEN the employee confirms the product was picked, THEN the system records it as picked.

AT-28B: GIVEN multiple products are required for an order, WHEN one product is confirmed as picked, THEN the remaining products remain unconfirmed.

AT-28C: GIVEN a product has already been confirmed as picked, WHEN the order is viewed again, THEN the product remains recorded as picked.

Dependencies / sequence: US-27 Prototype coverage: Not Represented

FINAL STORY 29 · US-30 · ROLE: Warehouse Employee · RANK: 12 · MOSCOW: M · MVP: N

Story

As a warehouse employee, I want to mark a completed warehouse order as ready for shipping so that the shipping process can continue.

Acceptance tests

AT-30a: GIVEN all required warehouse work for an order has been completed, WHEN the employee marks the order ready for shipping, THEN the system records the order as ready for shipping.

AT-30B: GIVEN required products have not all been successfully prepared, WHEN the employee attempts to mark the order ready for shipping, THEN the order is not recorded as ready for shipping.

AT-30C: GIVEN an order has been marked ready for shipping, WHEN the order is viewed afterward, THEN its status reflects that it is ready for shipping.

Dependencies / sequence: US-28 Prototype coverage: Not Represented

Change analysis and final reflection

Write 400–600 words total. Refer to specific story/test IDs, dependency entries, prototype evidence, and AI interaction IDs.

1 · AI draft quality

Describe at least one specific weakness or unsupported assumption in the initial AI draft. What did you change, and how did you know the change was needed?

One unsupported assumption made by the initial AI draft was that if an order's availability status was unknown, then it was not able to be ordered. This happens with two specific items, the pullover hoodie, and the tiger backpack. These two items get the same treatment as if they were unavailable, something never specified to do when creating the prototype. In tests AT-02a and AT-02B, there is no description of an unknown state. The only test that mentions an unknown state is AT-01C, which only says that it doesn't indicate that the items are available. It doesn't indicate anything that states that you can't purchase an item in an unknown state, just that it would be its own state outside of available or unavailable. I would add another test case that determines the definition of what an unknown state encapsulates. This change is needed as having a third state without a defined description could cause issues on what is to be assumed about it, instead of explicitly stated.

2 · Prioritization or dependency insight

Describe at least one change caused by ranking, MVP selection, or dependency analysis. Explain the tradeoff and why your final choice is defensible.

After deducing the final ranking, US-29 seemed to be misplaced or useless. When designing a system like this, the warehouse worker should not have to manually track or relay product outages. In a modern e-commerce architecture,

product outages and stock levels should be tracked automatically via integrated systems rather than relying on manual reporting by warehouse personnel. Requiring warehouse workers to manually log introduces human error, and adds more time required to complete an easily automated task. As a result, it was ideal to remove that generated user story. US-29 added unnecessary overhead without providing unique business value. This removal was not without tradeoffs. Our dependency analysis in Worksheet 4 had listed U-29 as a prerequisite for US-12. Removing US-29 broke that dependency, so the resolution was to point US-12's prerequisite to US-08, since sys admins already are supposed to maintain product availability data.

3 · Prototype insight

Describe at least one ambiguity, missing state, or testability issue discovered while planning or walking through the prototype. What changed in the prototype, backlog, or both?

A testability issue that we discovered in our prototype was that one of our acceptance tests was worded slightly ambiguously. It resulted in the test not really being able to occur because theoretically the unavailable items could never be added to orders. To fix this issue we reworded the acceptance test to more accurately fit the context and situations that could occur to ensure that our tests and prototype made more sense from a documentation standpoint. The prototype itself remained largely the same as did our backlog as it was a minor wording error that probably should have been caught earlier on in the worksheets.